

Course: Data Science for Clinicians

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Course Timeline: This is a two-day course, constituting 16 hours of material. There are pre-course materials that are designed for all students to complete prior to attending the course. These pre-course materials are designed for student success upon attendance and are highly recommended.

Course Rational: Current clinical medicine requires professionals who are confident and skilled in gathering, managing, analyzing, and interpreting data. This new clinical skillset requires data science core competencies including computer coding, data cleaning, data visualization, and data synthesis. However, current data science knowledge and skills in clinical training are sparse. This course enhances the medical curriculum by filling this clinical training gap where there is currently no formal training course.

Course Description: *Data Science for Clinicians* is designed to teach basic data science, epidemiological analyses, and clinical synthesis skills to clinical medical students and professionals to prepare them for modern medical practice. This course will be taught using open-access computer language software and publicly available datasets from different disciplines, to increase the transparency and sustainability of this program. This course is designed to couple knowledge and application to ensure participants can apply the course objectives. This curriculum provides the tools necessary to ask critical questions, training to think in data driven questions, and skills to synthesize individual patient and health system level results. This course will be taught with a mix of lectures and practicals covering the entire data science pipeline from data acquisition to presentation to clinical synthesis. The course will consist of two days of work.

Data Sets: All datasets used for this course are open source and freely accessible. They can be downloaded and shared, to explore these data. From experience, the more exploration and collaboration that occurs while learning will improve burgeoning data science skills.

- *Four Professional Leagues Data:* This is web scraped data from the four major professional sport leagues in North America. These data are injury surveillance and transactions data from 2007-2019. The data can be found here (under Supplement 1): <https://www.nature.com/articles/s41598-021-87920-6#Sec28>
- *National Basketball League (NBA) Data:* This is web scraped data of injury surveillance and performance data from the NBA from 2007-2020. The overall NBA data can be found here: <https://osf.io/xywq4/>. The NBA rookies data can be found here: https://github.com/gsb15/NBA_Rookies_Data.
- *'Running Through' Data:* These data are from a cohort of recreational runners from across Europe from 2020-2022, using weekly reporting, Garmin watch, and Strava, to obtain weekly running exposure and injuries. The data can be found here: <https://osf.io/jgfau/>.

Tools: There are specific computing and online tools required for this course. All are open access and freely accessible.

- *R Computer Software:* This course will be taught using R computer software. You can download the software here for MAC or PC: <https://www.r-project.org/>.
- *R Studio:* It is recommended to use R Studio as your GUI interface, instead of the basic R software. In my experience, it is easier, quicker, and more intuitive. You can download R studio for MAC or PC here: <https://posit.co/download/rstudio-desktop/>.
- *'Daggity':* Daggity is a free online tool to help visualize confounding relationships. This can be used online or downloaded to your computer. You can interface here: <https://www.daggity.net/>.

- *Kaggle*: Kaggle is an open access data repository, where you can find a plethora of different datasets. This is a great resource for exploring further data and honing your data science skills. You can find the repository here: <https://www.kaggle.com/datasets>.

Further Resources: Additional resources are at your fingertips for further reference. These are signposted for each learning module. All books are open source and free.

- Whickam H, Cetinkaya-Rundel M, Grolemund G (2023). *R for Data Science: Import, Tidy, Transform, Visualize, and Model Data* (2nd Edition). O'Reiley. <https://r4ds.hadley.nz/>
- Hernán MA, Robins JM (2020). *Causal Inference: What If*. Boca Raton: Chapman & Hall/CRC. <https://miguelhernan.org/whatifbook>
- UCLA Regression Tutorial: [https://stats.oarc.ucla.edu/wp-content/uploads/2022/11/R_reg_2022.html#\(1\)](https://stats.oarc.ucla.edu/wp-content/uploads/2022/11/R_reg_2022.html#(1))

Course Objectives/Outcomes:

At the completion of the course, students will:

1. Adhere strictly to ethical guidelines regarding participant consent, conflicts disclosure, data transparency and framing of patient projections communicated to key knowledge users.
2. Articulate methodology assumptions, strengths, and gaps transparently to allow independent expert scrutiny prior to supplying any data reports or analyses to clinicians, healthcare leaders, and/or researchers.
3. Visualize different data sources into best scientific practice for effective communication to clinicians, healthcare leaders, and/or researchers.
4. Identify, communicate, and control for confounding in clinical based questions.
5. Calculate prevalence, incidence, and rates and communicate the different clinical inferences of these epidemiological calculations.
6. Manage uncertainty bias, overfitting limitations, data leakage, and algorithmic bias to improve reliability and generalizability of patient data analyses.
7. Appraise best practices for prediction framework conceptualization spanning research inquiry, methodology, implementation, and responsible communication of patient projections.

Evaluation Tools Description:

Lectures & Readings: Lectures will commence throughout the course. Your course booklet will parallel the lectures and course assignments.

Assignments: Throughout the course, you will use these open data to perform different assignments following each lecture. From experience, using exploring, analyzing, and collaborating on real-world data is the best teacher when becoming a data scientist.

Pre-Course Lectures and Materials

To prepare for successful completion of this course, it is recommended to have a basic knowledge of coding skills using the R statistical programming language. If you have no or minimal experience in this language, it is highly recommended to engage and to continue to reference the pre-course materials.

Topic	Course Material	Assignments
Downloading and opening R and R Studio	https://www.r-project.org/ https://posit.co/download/rstudio-desktop/	Online videos and practicals
Installing R Tools	https://cran.r-project.org/bin/windows/Rtools/	Online videos and practicals
Installing packages		Online videos and practicals
Downloading data from Kaggle	https://www.kaggle.com/datasets .	Online videos and practicals
Uploading data to R		Online videos and practicals
Using packages and basic commands in R		Online videos and practicals
Data and code workflow basics		Online videos and practicals
Tidy data		Online videos and practicals

Course Schedule:

Lecture	Topic(s)	Course Materials (text chapters, scientific papers, etc.)	Aligned Course Objective(s) *Number only	Assignment(s)
0	• Introduction	Pre-course materials		
1	• Protected Health Information • Ethical Consideration • Open Science	https://www.hhs.gov/hipa/a/for-professionals/index.html Collins et al. TRIPOD+ AI statement: updated guidance for reporting clinical prediction models that use regression or machine learning methods. <i>bmj</i>. 2024 Apr 16;385 Bullock GS et al. Call for open science in sports medicine. <i>British Journal of Sports Medicine</i>. 2022 Oct 1;56(20):1143-4. Bullock GS et al. Paving the way for greater open science in sports and exercise medicine: navigating the barriers to adopting open and accessible data practices. <i>British Journal of Sports Medicine</i>. 2024 Mar 1;58(6):293-5.	1	Using the NBA data, de-identify, De-personalize & pseudonmize the data.
2	• Data transformation • Data joining	R for Data Science Chapters 12, 13, 14, 19	2	Using the NBA data, transform the days missed data from a string to numeric. Calculate total days missed from the

				game data. Join the data with another NBA data set.
3	- Cleaning 'Dirty' Data - Missing Data (brief) - Data Leakage	R for Data Science Chapters: 5	2	Using the Four Professional Leagues Data, clean the data, ready for further work.
4	Best scientific practice data visualization	R for Data Science Chapters 1, 9, 10, 11	3	Create data visualization reports with the 'Running Through' data for use with 1. Patient 2. Clinician 3. Health Leader
5	Missing data	Altman et al. Missing Data. BMJ. 2007;334(7590):424. Lee et al. Framework for the treatment and reporting of missing data in observational studies: the treatment and reporting of missing data in observational studies framework. J Clin Epidemiol. 2021:79-88.	2	Using the <i>nanian</i> package, evaluate the NBA data for missing data prevalence, patterns, and mechanisms. Create visualizations using different groupings of variables and also the shadow matrix.
6	- Prevalence - Incidence - Rate	Knowles SB, Marshall SW, Guskiewicz KM. Issues in estimating risks and rates in sports injury research. Journal of athletic training. 2006;41(2):207. Hopkins WG, Marshall SW, Quarrie KL, Hume PA. Risk factors and risk statistics for sports injuries. Clinical Journal	5	Using the NBA data, calculate prevalence, incidence, and rates for each position and for ankle, knee, and low back injuries. Create a results report to inform a healthcare leader (see assignments below).

		of Sport Medicine. 2007 May 1;17(3):208-10.		
7	• Confounding • Diagram Acyclic Graphs (DAG's) • Collider Bias	<i>Causal Inference: What If:</i> Chapters 1, 6, 7	4	Using the online ' <i>DAGGITY</i> ' (https://www.dagitty.net/) web application, create a DAG of a cause-effect clinical relationship of your choosing.
8	• Regression • Developing a regression model • Different forms of generalized linear models • Interpreting regression models	UCLA Regression Tutorial Myers RH, Montgomery DC. A tutorial on generalized linear models. <i>Journal of Quality Technology</i> . 1997 Jul 1;29(3):274-91. DeMaris A. A tutorial in logistic regression. <i>Journal of Marriage and the Family</i> . 1995 Nov 1;956-68.	7	Using the Running Through data, develop a 1) linear and 2) logistic regression to evaluate the association of 1) running distance and speed; 2) running distance and lower extremity injury
9	• What actually is prediction? • Basic steps in developing a prediction model	Bullock et al. Methods matter: clinical prediction models will benefit sports medicine practice, but only if they are properly developed and validated. <i>British journal of sports medicine</i> . 2021 Dec 1;55(23):1319-21. Bullock et al. Clinical prediction models in sports medicine: a guide for clinicians and researchers. <i>journal of orthopaedic & sports</i>	7	Within the NBA data, develop a prediction model to forecast injury.

		physical therapy. 2021 Oct;51(10):517-25. Rhon et al. Predictive models for musculoskeletal injury risk: why statistical approach makes all the difference. BMJ Open Sport & Exercise Medicine. 2022 Oct 1;8(4):e001388.		
10	Putting your new skills into practice.		6	